

التغذية العلاجية  
والسمنة



أكاديمية مصر الدولية



أكاديمية مصر الدولية ترحب بكم

اضغط هنا..

# **Non surgical approaches for fat reduction ?**

**Yasser A. Hamed M.D.**  
**Al-Azhar university**

**Research Associate Minneapolis University Minnesota USA**  
**[lasercairo@yahoo.com](mailto:lasercairo@yahoo.com)**

# *History of Mesotherapy*

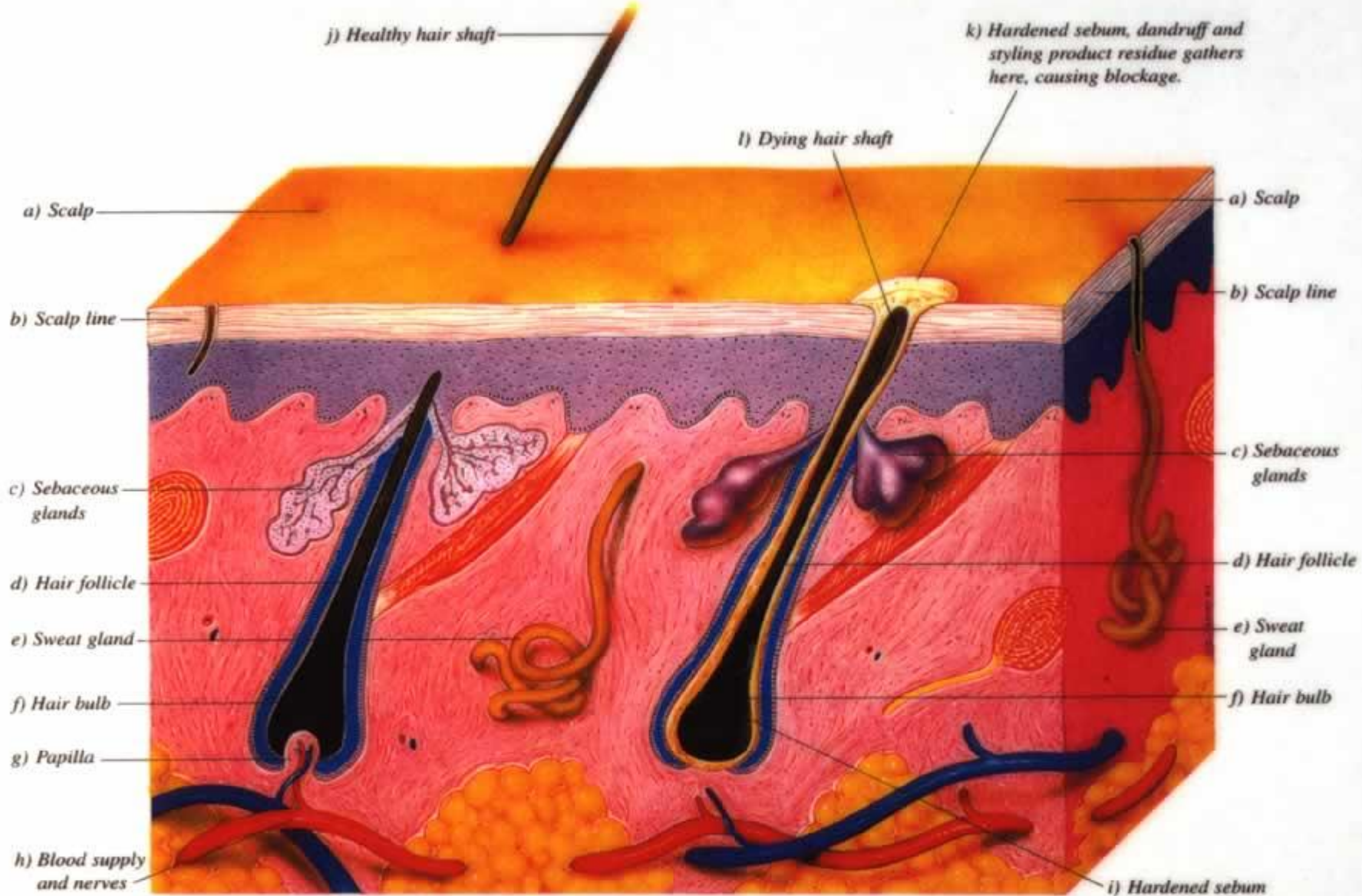


- Dr. Michel Pistor (1924-2003) performed clinical research and founded the field of mesotherapy.
- The French press coined the term Mesotherapy in 1958.
- The French Academy of Medicine recognized Mesotherapy as a Specialty of Medicine in 1987.
- 18,000 physicians worldwide.

# Mesotherapy

- **Mesotherapy** (from Greek mesos, "middle", and therapy from Greek therapeia, "to treat")
- Multiple non-surgical injections of pharmaceutical and homeopathic medications, plant extracts, vitamins, and other ingredients.





**HEALTHY FOLLICLE WITH SEBUM  
FLOWING NORMALLY**

**TYPICAL BLOCKED FOLLICLE**

# *Indications of Mesotherapy*

- Lipolysis and treating cellulite
- Telogen effluvium, and male pattern alopecia
- Hyperpigmentary disorders
- Skin rejuvenation

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# TECHNIQUE OF INJECTIONS

- POINT TO POINT TECHNIQUE or STEP BY STEP TECHNIQUE
- 13 mm depth
- 0.5 cc per fraction.
- Space every 2 cm
- FREQUENCY:
- 1 SESSION EVERY 2 TO 3 WEEKS



# TECHNIQUE OF INJECTIONS

- Manually
- Meso-gun
- Trans epidermal delivery



- Point to point technique
- Nappage technique



# ***TECHNIQUE OF INJECTIONS***

- Inject 0.5 cc per fraction
- Space every centimeter



# TECHNIQUE OF INJECTIONS

## ■ Meso-gun



# TECHNIQUE OF INJECTIONS

- Syringe injection
- Multiple injectors



# Phosphatidylcholine

- Phosphatidylcholine is the most important essential phospholipids in the human body.
- It facilitates the emulsification of fat into tiniest particles, enabling the absorption and transportation of fat then the adipocytes burst.

# Mechanism of action

- phosphatidylcholine in fatty tissue is hydrolyzed through phospholipase A2 and D, resulting in hydrolyzing triglycerides to diglycerides and then to monoglycerides, ----- fatty acids and glycerol ----- the bloodstream by lymphatics.



# Phosphatidylcholine and Deoxycholate

Aesthet Surg J. Mar 2013

## Metabolic and Structural Effects of Phosphatidylcholine and Deoxycholate Injections on Subcutaneous Fat

PC-DC injections can effectively reduce abdominal fat volume and thickness by inducing adipocyte necrosis. These treatments do not increase circulating markers of inflammation or affect glucose and lipid metabolism.

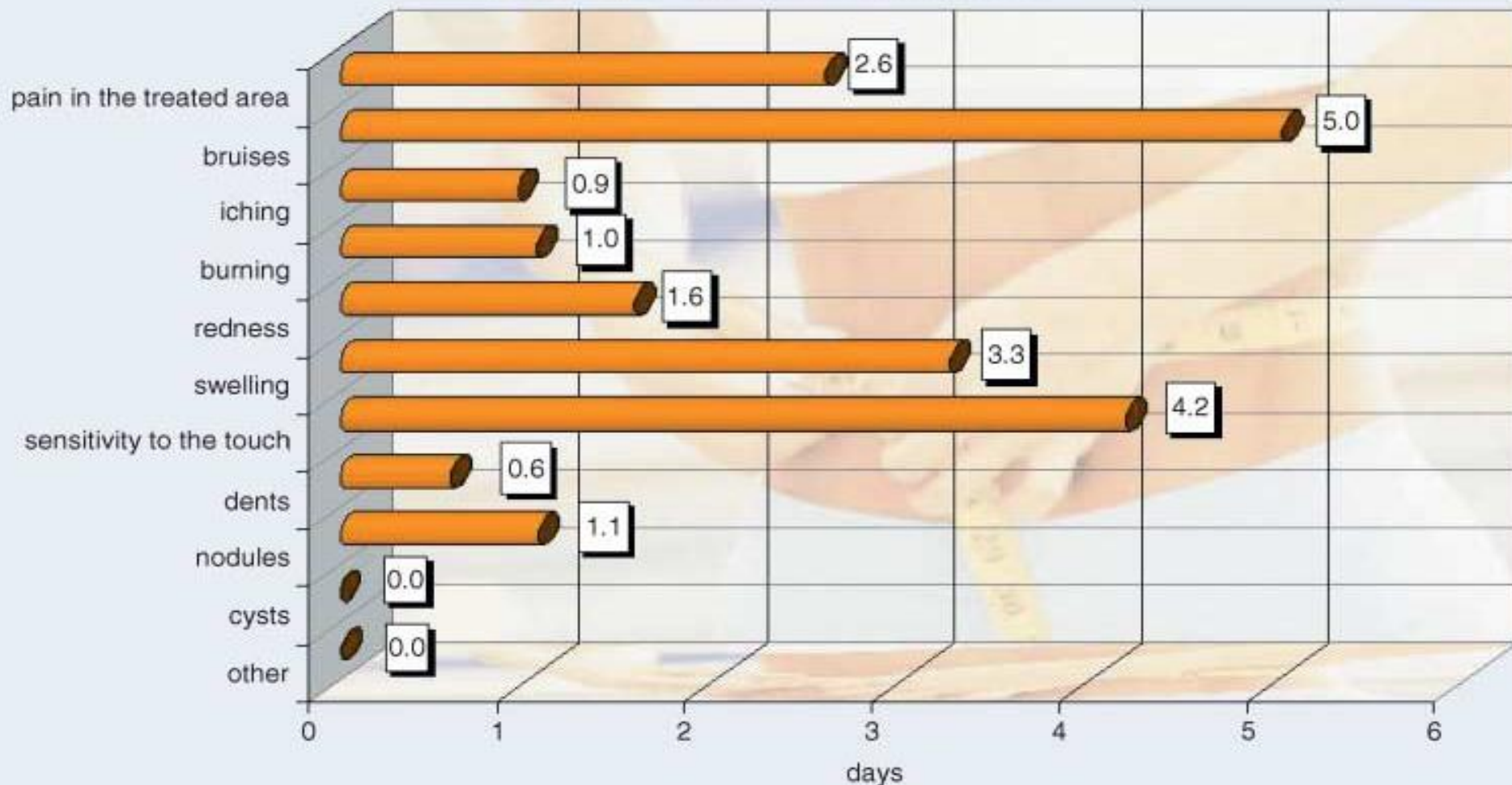
J Eur Acad Dermatol Venereol. 2014

Efficacy, patient-reported outcomes and safety profile of (deoxycholic acid), an injectable drug for the reduction of unwanted submental fat: results from a phase III, randomized, placebo-controlled study.

**CONCLUSION:** deoxycholic acid was effective well tolerated, and may be an alternative to surgery for improvement of their submental profile.

# Side effects

side effects - average duration over all regions  
out of 753 treatments



# *Advantage of Mesotherapy*

- Simple
- Non-invasive
- Effective
- Minimal side effects ?



# ***Mesotherapy side effects***

***Mesotherapy side effects***



# Mesotherapy side effects

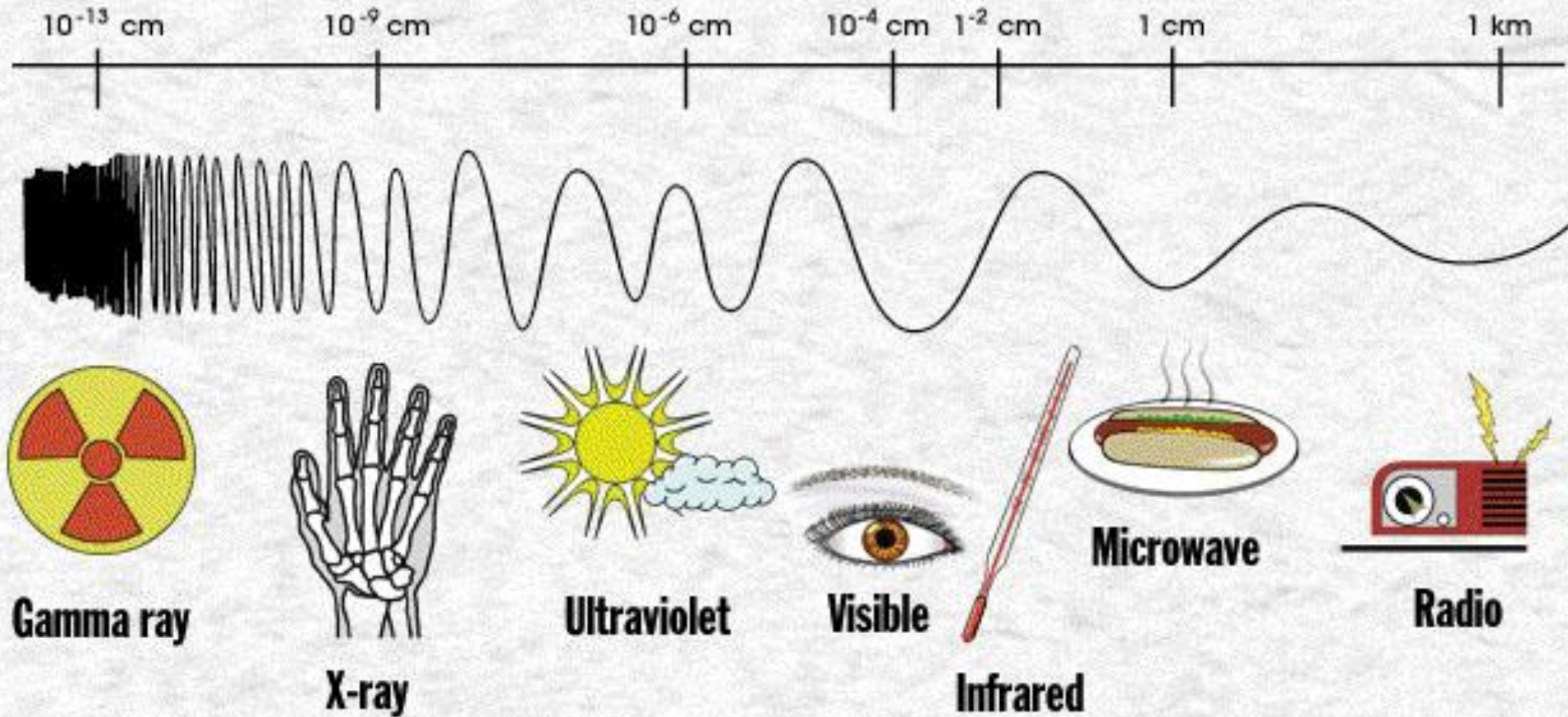
- 1: Mesotherapy-induced urticaria. *Dermatol Surg.* 2010 Aug
- 2: Cutaneous tuberculosis after mesotherapy: report of six cases. *Biomedica.* 2010
- 3: A case of the Behcet's disease diagnosed by the panniculitis after mesotherapy. *Rheumatol Int.* 2010
- 4: Ischemic colitis after mesotherapy combined with anti-obesity medications. *World J Gastroenterol.* 2010
- 5: Three cases of panniculitis due to Mycobacterium abscessus after mesotherapy. *Actas Dermosifiliogr.* 2010

# *Mesotherapy side effects*

- Nontuberculous mycobacteria infection after mesotherapy: preliminary report of 15 cases  
*Int J Dermatol. 2007 Jun;46(6):649-53*
- Abdominal haematoma--a mesotherapy complication  
*Acta Derm Venereol. 2005;85(5):446.*
- Localized urticaria pigmentosa triggered by mesotherapy.  
*Dermatology. 2004;209(4):343-4*

***Radio frequency RF***  
**Radio frequency RF**

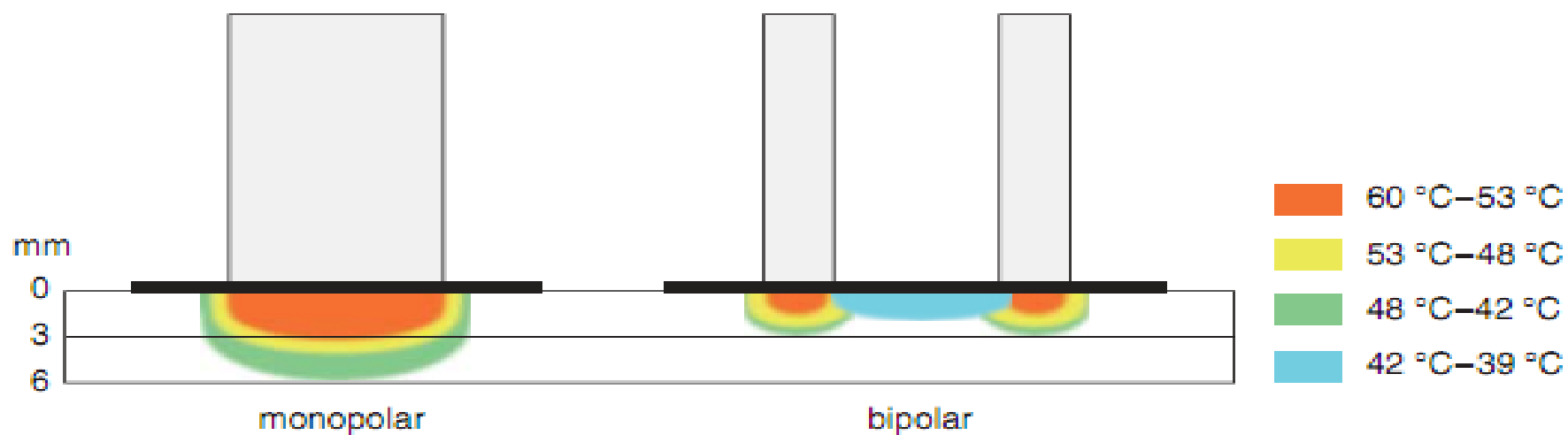
# *The Electromagnetic Spectrum*



# Types of Radio Frequency (RF)

- Mono polar
- Bipolar
- Needle or Fractional









Small Probe for Face

Big Probe for body/neck

Tripolar For body/Jowl



bbsalonequipment

# High Focused Ultrasound

*Focused ultrasound  
therapeutic  
transducer*

*Low energy  
everywhere except in  
the focal area*



The image features a solid green background. A white rounded rectangle is positioned in the upper left quadrant. The text "Dermaroller™" is centered within this white shape. Below the white shape, a dark blue horizontal bar with rounded ends extends across the lower right portion of the image.

**Dermaroller™**

**Dramatic increase of new collagen fibers up to 1,000% in some cases.**

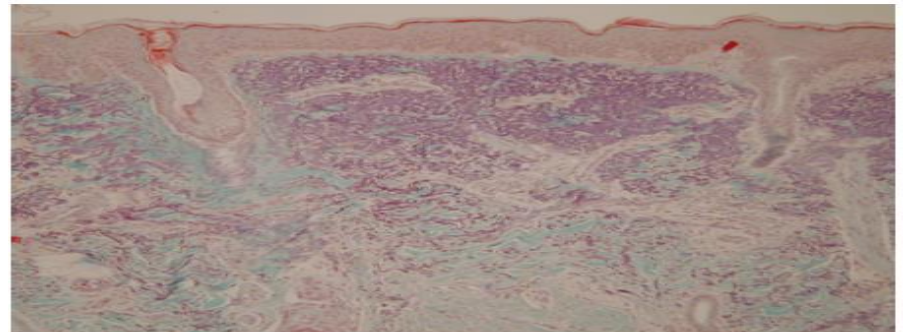
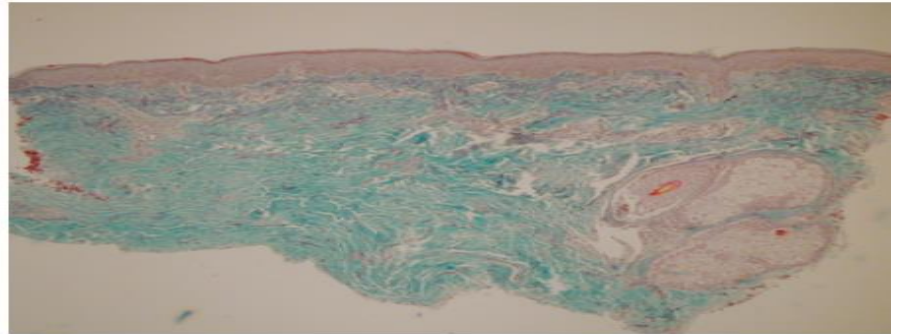
Histologic study



# Mechanism of action

- This was proven by histological pictures.
- Biopsy before needling→
- Biopsy after needling (collagen fibers stained violet)→

Biopsy before needling

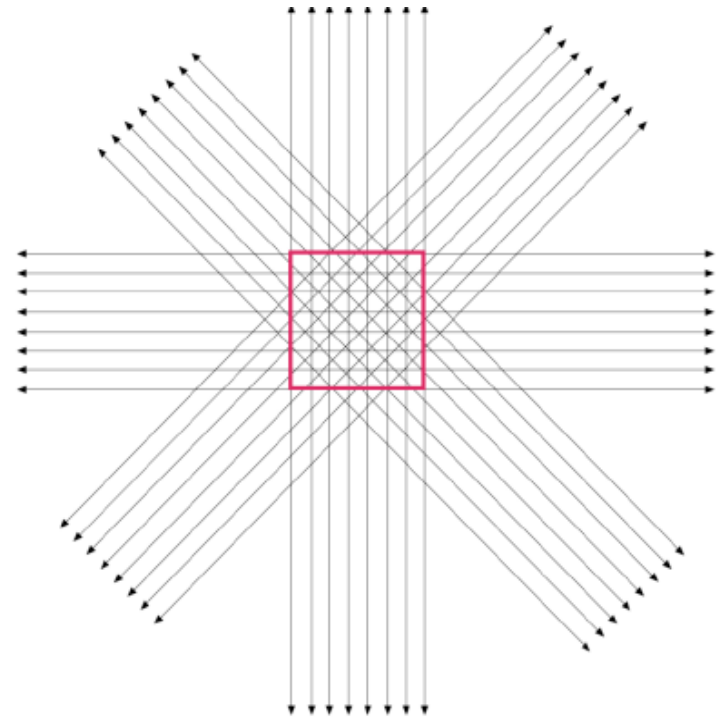


Biopsy after needling (collagen fibers stained violet)



# procedure

- Roll each part 4-5 times in 4 directions: vertically ,horizontally , and in both diagonals



# Dermaroller

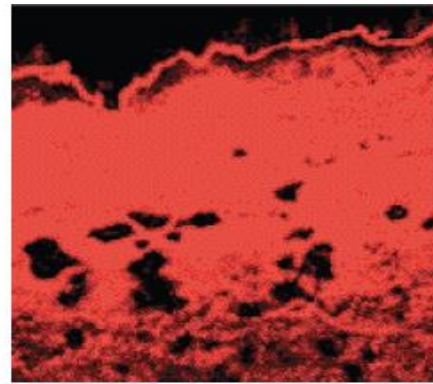
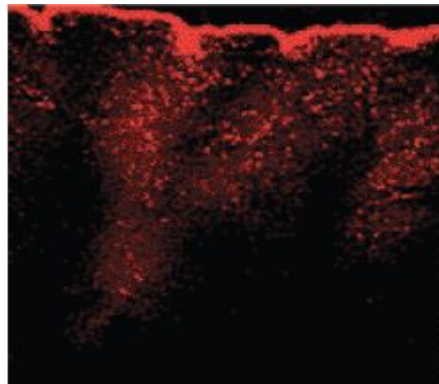
Trans dermal delivery  
system



# Histology proof

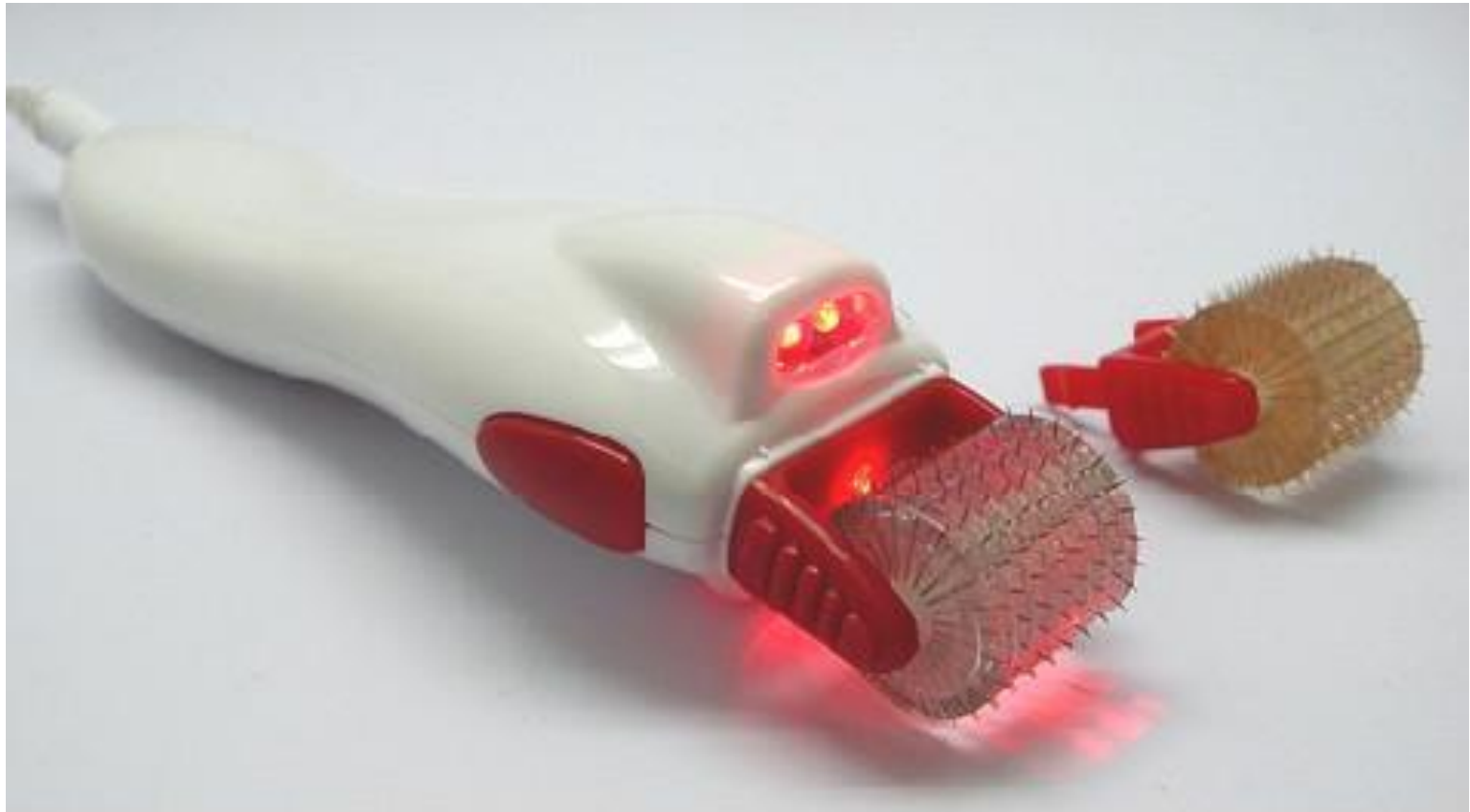
- This was proven by histological pictures :  
Comparing between normal creaming and cream plus dermarollar .

Findings of university of Marburg/Germany,2000.



# Different types of Rollers:

LED Photon derma roller



# Different types of Rollers:

**Derma Roller Auto Needle**



# Different types of Rollers:

Derma Pen







**Thank you**